3D façades for Ktirio Urban Building

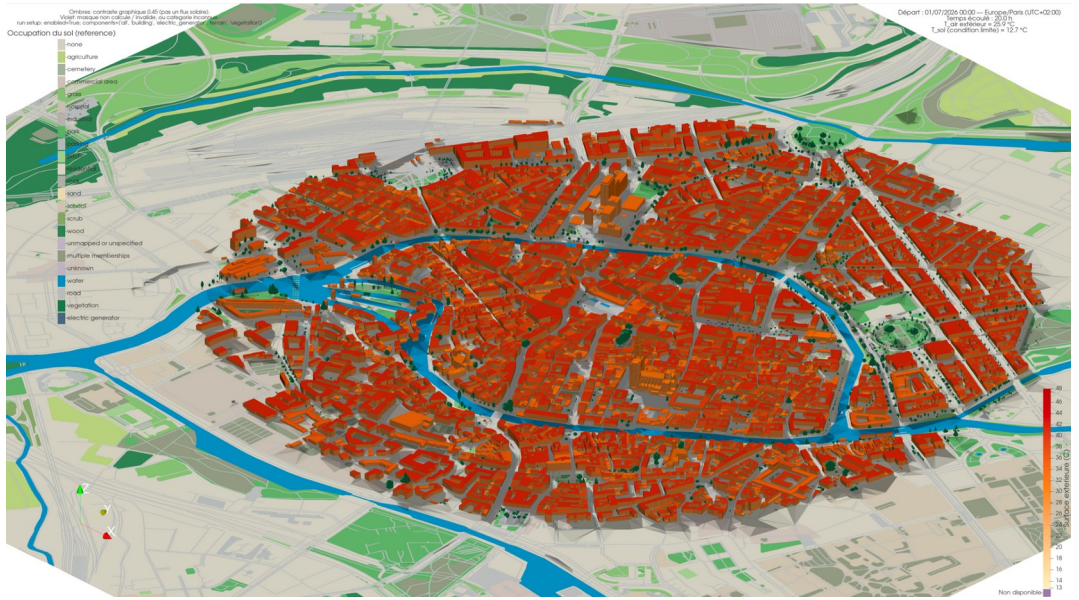
Ktirio reconstructs cities from OSM and enriches attributes with BDNB.

Existing context: terrain, land use, water and mapped vegetation.

Strasbourg city reconstruction



Later view with shadows



Project: enrich façades and check missing information in the surroundings.

Project objectives

Identify and retrieve the correct building

Reconcile OSM, BDNB and available RNB links. Verify the footprint and source tiles.

Recover missing envelope and context

Estimate windows, window-to-wall ratio (WWR), heights and storeys. Retrieve vegetation and nearby geometry.

Validate and export to Ktirio

Compare with NEPSEN references. Retain provenance, missing data and quality indicators.

Apave · Artigues-près-Bordeaux



Partner site plan

Renovated site

Connected wings and a landscaped courtyard.

Reference material

Site and floor plans.
Exterior views and usage review.

Validation task

Match the correct wings and renovated façade geometry.

Data gate: geometry and imagery must match the renovated envelope.

Téthys · Bègles



Partner exterior view

Recent office construction

Parc Newton, Bègles.
Realization reviewed in 2023.

Reference material

Floor plans, section and glazing data.
Exterior views.

Validation task

Recover windows and storey bands.
Handle shutters and vegetation.

Data gate: the Bordeaux 2012 textured model predates this project.

Haut Blanzac · Mérignac



Existing building · partner reference

Housing renovation

Envelope upgrades and changes to the surrounding vegetation.

Reference material

Existing and proposed floor plans. Conception review from 2024.

Validation task

Compare windows, heights and vegetation for the chosen state.

Data gate: distinguish the existing envelope from the renovation design.

Mathematical challenges

Problem	Mathematical approach
Building correspondence	Polygon overlap and spatial consistency. Explicit handling of split or grouped buildings.
Façade reconstruction	Robust plane fitting and coordinate transforms. Projection onto a metric façade frame.
Window and façade areas	Segmentation masks, geometric area calculations. WWR = window area / gross façade area.
Incomplete observations	Occlusion masks and missing values. Error analysis against annotated references.

Each value must be labelled as measured, estimated or unavailable.

Computing and engineering challenges

Challenge	Concrete project task
Data engineering	Reconcile IDs, coordinates, source dates and tiles. Select the target building and its surroundings.
Computer vision	Detect windows in imperfect façade textures. Handle trees, shutters, balconies and shadows.
Software and performance	Repeatable runs, cached data, bounded memory. Measure processing time and export traceable inputs.
Engineering validation	Verify the building against independent references. Test wrong neighbours, tile edges and missing data.

First acceptance test: the intended building and required context load correctly.

October–January roadmap

Period	Milestone
October	Building identity and data retrieval. Recovery test for the target and its context.
November	Metric façade reconstruction. Window annotations and reference conventions.
December	Window extraction and envelope estimates. Ktirio export with provenance and quality.
January	Accuracy, coverage and runtime evaluation. Documented code, report and oral presentation.

One student, a few hours per week. NEPSEN cases or a Strasbourg fallback.

Christophe Prud’homme · CEMOSIS / IRMA · In collaboration with NEPSEN

Annex · Dataset URLs

OpenStreetMap · building footprints

https://www.openstreetmap.org/

BDNB · building attributes

https://bdnb.io/

RNB · national building identifiers

https://rnb.beta.gouv.fr/

Strasbourg · textured mesh (2022)

https://data.strasbourg.eu/explore/dataset/pm3d_2022/information/

Strasbourg · 3D model + trees (2022)

https://data.strasbourg.eu/explore/dataset/odata3d_maquette_2022/information/

Bordeaux · textured buildings (2012)

https://datahub.bordeaux-metropole.fr/explore/dataset/bati3d/

Bordeaux · LiDAR point cloud (2020)

https://datahub.bordeaux-metropole.fr/explore/dataset/met_nuage_points_lidar_2020/

Bordeaux · surface model (2023)

https://datahub.bordeaux-metropole.fr/explore/dataset/met_mns_2023/

IGN · classified LiDAR HD

https://cartes.gouv.fr/rechercher-une-donnee/dataset/IGNF_NUAGES-DE-POINTS-LIDAR-HD